

Lecture Notes

EC226 Term 2 2024/25

Serial Correlation

Readings:

- Stock and Watson (2003): Chapter 14.3
- Dougherty (2016): Chapter 12
- Wooldridge (2013): Chapter 11

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1 Introduction

Serial correlation for a series, z_t , is the presence of some form of linear dependence over time. The AutoCorrelation Function (ACF) is a pictorial representation of this linear dependency, which is measured in the form of a correlation between z_t and z_{t-k} for different values of k , that is:

$$\text{corr}(z_t, z_{t-k}) = \frac{\text{cov}(z_t, z_{t-k})}{\sqrt{V(z_t)V(z_{t-k})}} = \frac{\text{cov}(z_t, z_{t-k})}{V(z_t)} = \frac{\gamma_k}{\gamma_0} = \rho_k$$

where $\rho_0 = 1$ and $|\rho_j| < 1$, $j \geq 1$. We will look at the ACF for four types of models

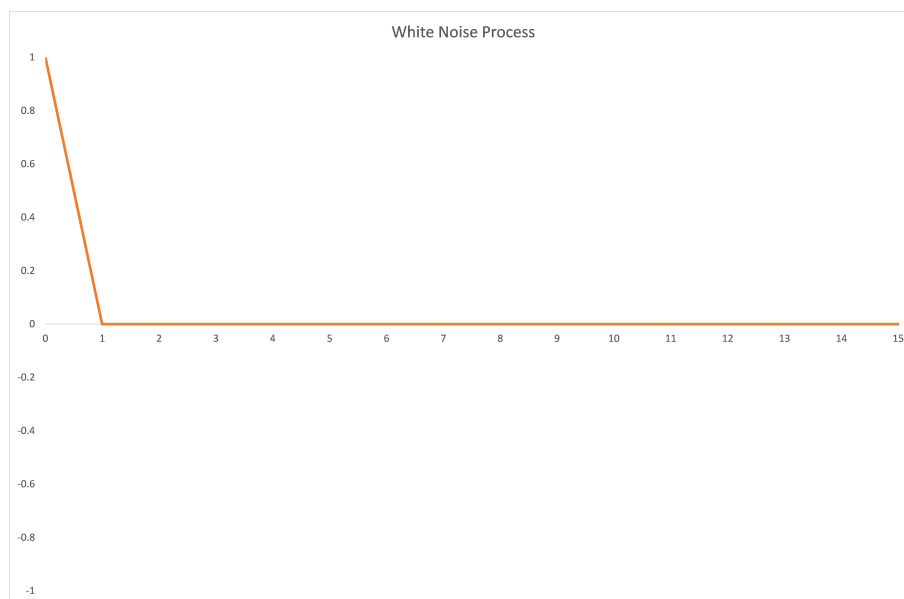
1. white noise
2. AutoRegressive (AR)
3. Moving Average (MA)
4. AutoRegressive Moving Average (ARMA)

2 White Noise

$$z_t = \varepsilon_t \quad \varepsilon_t \sim N(0, \sigma^2)$$

The process z_t , is called a White Noise process and has the same mean, the same variance and is not correlated over time, i.e. $E(z_t) = 0$; $V(z_t) = \sigma_z^2$; and $cov(z_t, z_{t-j}) = 0, j > 0 \Rightarrow \rho_k = 0$, for $k \geq 1$ and the theoretical ACF is in Figure 1 (see Appendix A Figure A1 for a sample plot of a white noise process and the associated sample ACF and PACF. Appendix B (C) gives the Stata (R) code to generate some of these ARMA processes.

Figure 1



3 AutoRegressive (AR) Model

3.1 AR(1)

$$z_t = \phi z_{t-1} + \varepsilon_t \quad \varepsilon_t \sim N(0, \sigma^2)$$

where $|\phi| < 1$ (means the process is stationary).

The mean of an AR(1) model is calculated as:

$$\begin{aligned} E(z_t) &= E(\phi z_{t-1} + \varepsilon_t) \\ &= E(\phi z_{t-1}) + E(\varepsilon_t) \\ &= \phi E(z_{t-1}) \text{ (as } E(\varepsilon_t) = 0) \\ &= \phi E(z_t) \text{ (due to stationarity)} \\ (1 - \phi)E(z_t) &= 0 \end{aligned}$$

To derive the ACF for the AR model we derive the Yule-Walker equations:

$$\begin{aligned} V(z_t) = \gamma_0 &= E(z_t z_t) = E[(\phi z_{t-1} + \varepsilon_t) z_t] = \phi E(z_{t-1} z_t) + E(z_t \varepsilon_t) \\ \gamma_0 &= \phi \gamma_1 + \sigma^2 \end{aligned} \tag{1}$$

$$\begin{aligned} \text{cov}(z_t, z_{t-1}) = \gamma_1 &= E(z_t z_{t-1}) = E[(\phi z_{t-1} + \varepsilon_t) z_{t-1}] = E(\phi z_{t-1}^2) + E(\varepsilon_t z_{t-1}) = \phi E(z_{t-1}^2) \\ \gamma_1 &= \phi V(z_t) \Rightarrow \gamma_1 = \phi \gamma_0 \end{aligned} \tag{2}$$

$$\begin{aligned} \text{cov}(z_t, z_{t-2}) = \gamma_2 &= E(z_t z_{t-2}) = E[(\phi z_{t-1} + \varepsilon_t) z_{t-2}] = E(\phi z_{t-1} z_{t-2}) + E(\varepsilon_t z_{t-2}) = \phi E(z_{t-1} z_{t-2}) \\ \gamma_2 &= \phi \gamma_1 \end{aligned} \tag{3}$$

Solving (1) and (2) we get:

$$\gamma_0 = \phi^2 \gamma_0 + \sigma^2 \Rightarrow \gamma_0 = \sigma^2 / (1 - \phi^2)$$

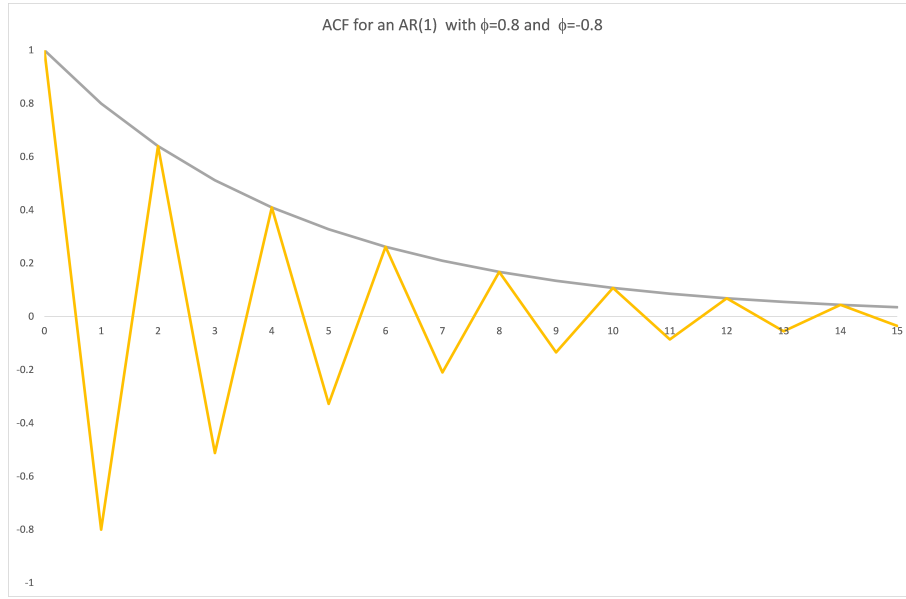
and $\gamma_1 = \phi \sigma^2 / (1 - \phi^2)$ and recursively $\gamma_j = \phi^j \sigma^2 / (1 - \phi^2)$ $j \geq 1$

In which case:

$$\begin{aligned} \rho_1 &= \frac{\gamma_1}{\gamma_0} = \frac{\phi V(z_t)}{V(z_t)} = \phi \\ \rho_2 &= \frac{\gamma_2}{\gamma_0} = \frac{\phi \gamma_1}{\gamma_0} = \phi \rho_1 = \phi^2 \\ \rho_3 &= \frac{\gamma_3}{\gamma_0} = \frac{\phi \gamma_2}{\gamma_0} = \frac{\phi^2 \gamma_1}{\gamma_0} = \phi^3 \\ \rho_j &= \phi^j \end{aligned}$$

See Figure 2 for the theoretical ACF for an AR(1) (Appendix A Figure A2 and Figure A3 provide sample plots of an AR(1) process and the associated sample ACF and PACF)

Figure 2



We can write our AR(1) in terms of lags of ε_t , define the lag operator, L , such that $Lz_t = z_{t-1}$, and $L^j z_t = z_{t-j}$, in which case we can write our AR(1) model as:

$$z_t = \phi L z_t + \varepsilon_t \Rightarrow z_t(1 - \phi L) = \varepsilon_t \Rightarrow z_t = (1 - \phi L)^{-1} \varepsilon_t$$

As $(1 - \phi L)^{-1} = 1 + \phi L + \phi^2 L^2 + \phi^3 L^3 + \dots$, we can write:

$$z_t = (1 + \phi L + \phi^2 L^2 + \phi^3 L^3 + \dots) \varepsilon_t = \varepsilon_t + \phi \varepsilon_{t-1} + \phi^2 \varepsilon_{t-2} + \phi^3 \varepsilon_{t-3} + \dots$$

and we have written our AR(1) model as an MA(∞) (see below for an introduction to Moving Average (MA) models).

Given the above form for the AR(1) written as an MA(∞) we can derive the mean, variance and autocovariances as:

$$E(z_t) = E(\varepsilon_t) + \phi E(\varepsilon_{t-1}) + \phi^2 E(\varepsilon_{t-2}) + \dots = 0$$

$$V(z_t) = V(\varepsilon_t) + \phi^2 V(\varepsilon_{t-1}) + \phi^4 V(\varepsilon_{t-2}) + \dots = \sigma^2 / (1 - \phi^2)$$

$$E(z_t, z_{t-1}) = E[(\varepsilon_t + \phi \varepsilon_{t-1} + \phi^2 \varepsilon_{t-2} + \dots)(\varepsilon_{t-1} + \phi \varepsilon_{t-2} + \phi^2 \varepsilon_{t-3} + \dots)]$$

$$E(z_t, z_{t-1}) = \phi E(\varepsilon_{t-1}^2) + \phi^3 E(\varepsilon_{t-2}^2) + \phi^5 E(\varepsilon_{t-3}^2) + \dots = \phi \sigma^2 / (1 - \phi^2)$$

3.2 AR(2)

$$z_t = \phi_1 z_{t-1} + \phi_2 z_{t-2} + \varepsilon_t \quad \varepsilon_t \sim N(0, \sigma^2)$$

and the process is assumed to be stationary, so $E(z_t) = E(z_{t-j})$ and $V(z_t) = V(z_{t-j})$.

The mean of the AR(2) is calculated as:

$$\begin{aligned} E(z_t) &= E(\phi_1 z_{t-1} + \phi_2 z_{t-2} + \varepsilon_t) \\ &= E(\phi_1 z_{t-1}) + E(\phi_2 z_{t-2}) + E(\varepsilon_t) \\ &= \phi_1 E(z_{t-1}) + \phi_2 E(z_{t-2}) \\ &= \phi_1 E(z_t) + \phi_2 E(z_t) \\ &= (1 - \phi_1 - \phi_2)E(z_t) = 0. \end{aligned}$$

In this case, the Yule-Walker equations can be derived as follows:

$$\begin{aligned} V(z_t) &= \gamma_0 = E(z_t z_t) \\ &= E[(\phi_1 z_{t-1} + \phi_2 z_{t-2} + \varepsilon_t) z_t] \\ &= \phi_1 E(z_{t-1} z_t) + \phi_2 E(z_{t-2} z_t) + E(\varepsilon_t z_t) \\ \gamma_0 &= \phi_1 \gamma_1 + \phi_2 \gamma_2 + \sigma^2 \end{aligned} \tag{4}$$

$$\begin{aligned} cov(z_t, z_{t-1}) &= \gamma_1 = E(z_t \cdot z_{t-1}) \\ &= E[(\phi_1 z_{t-1} + \phi_2 z_{t-2} + \varepsilon_t) z_{t-1}] \\ &= \phi_1 E(z_{t-1}^2) + \phi_2 E(z_{t-2} z_{t-1}) + E(\varepsilon_t z_{t-1}) \\ \gamma_1 &= \phi_1 \gamma_0 + \phi_2 \gamma_1 \end{aligned} \tag{5}$$

$$\begin{aligned} cov(z_t, z_{t-2}) &= \gamma_2 = E(z_t \cdot z_{t-2}) = E[(\phi_1 z_{t-1} + \phi_2 z_{t-2} + \varepsilon_t) z_{t-2}] \\ &= \phi_1 E(z_{t-1} z_{t-2}) + \phi_2 E(z_{t-2}^2) + E(\varepsilon_t z_{t-2}) \\ \gamma_2 &= \phi_1 \gamma_1 + \phi_2 \gamma_0 \end{aligned} \tag{6}$$

$$\begin{aligned} cov(z_t, z_{t-3}) \gamma_3 &= E(z_t \cdot z_{t-3}) = E[(\phi_1 z_{t-1} + \phi_2 z_{t-2} + \varepsilon_t) z_{t-3}] \\ &= \phi_1 E(z_{t-1} z_{t-3}) + \phi_2 E(z_{t-2} z_{t-3}) + E(\varepsilon_t z_{t-3}) \\ \gamma_3 &= \phi_1 \gamma_2 + \phi_2 \gamma_1 \end{aligned} \tag{7}$$

And by extension:

$$\gamma_j = \phi_1 \gamma_{j-1} + \phi_2 \gamma_{j-2} \quad j \geq 2 \tag{8}$$

dividing both side by γ_0 we have:

$$\rho_j = \phi_1 \rho_{j-1} + \phi_2 \rho_{j-2} \quad j \geq 2$$

which implies:

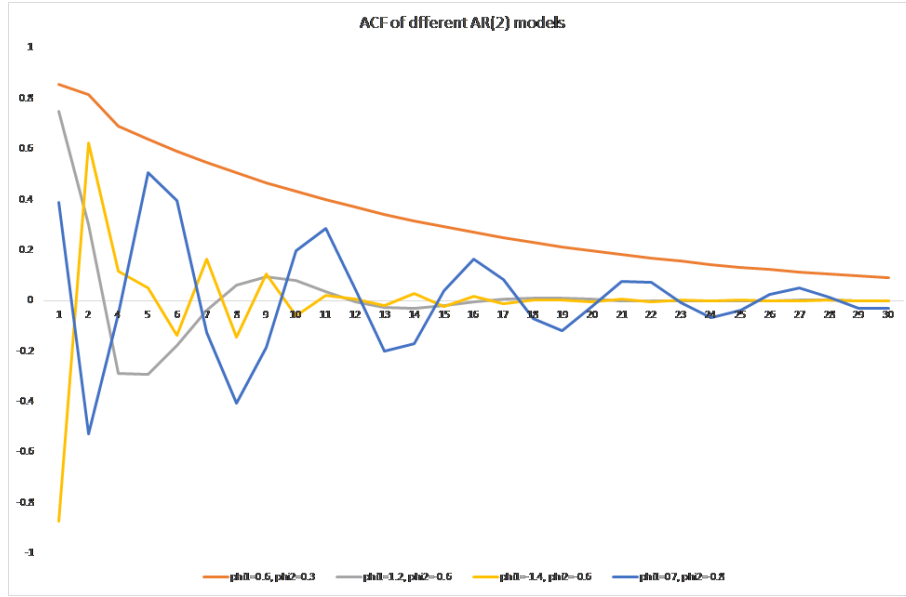
$$\rho_1 = \phi_1 \rho_0 + \phi_2 \rho_{-1}$$

and solving for ρ_1 we have

$$\rho_1 = \phi_1 / (1 - \phi_2)$$

and the higher order values can then be solved recursively. Figure 3 plots the theoretical ACF for an AR(2) for different values of ϕ_1 and ϕ_2 . Again this model could be rewritten as an MA(∞) but it is much harder to derive the mean, variance and autocovariances from this representation of the process.

Figure 3



3.3 AR(p)

$$z_t = \phi_1 z_{t-1} + \phi_2 z_{t-2} + \dots + \phi_p z_{t-p} + \varepsilon_t$$

where ε_t is a white noise process and the process is assumed to be stationary. In this case the Yule-Walker equations can be written:

$$V(z_t) = \gamma_0 = \phi_1 \gamma_1 + \phi_2 \gamma_2 + \phi_3 \gamma_3 + \dots + \phi_p \gamma_p + \sigma^2$$

$$\gamma_1 = \phi_1 \gamma_0 + \phi_2 \gamma_1 + \phi_3 \gamma_2 + \dots + \phi_p \gamma_{p-1}$$

$$\gamma_2 = \phi_1 \gamma_1 + \phi_2 \gamma_0 + \phi_3 \gamma_1 + \dots + \phi_p \gamma_{p-2}$$

...

$$\gamma_j = \phi_1 \gamma_{j-1} + \phi_2 \gamma_{j-2} + \dots + \phi_p \gamma_{j-p} \quad j \geq p$$

and therefore dividing both sides by γ_0 we get:

$$\rho_j = \phi_1 \rho_{j-1} + \phi_2 \rho_{j-2} + \dots + \phi_p \rho_{j-p}$$

3.4 Calculating Roots

For an AR(p) process there are p roots. These can be calculated by solving the p^{th} order polynomial

$$w^p - \phi_1 w^{p-1} - \phi_2 w^{p-2} - \dots - \phi_p = 0$$

So for an AR(1), this involves solving the equation:

$$w - \phi_1 = 0$$

which implies there is one root, $w = \phi_1$ and stationarity requires this root to be < 1 .

For an AR(2) one is solving the equation

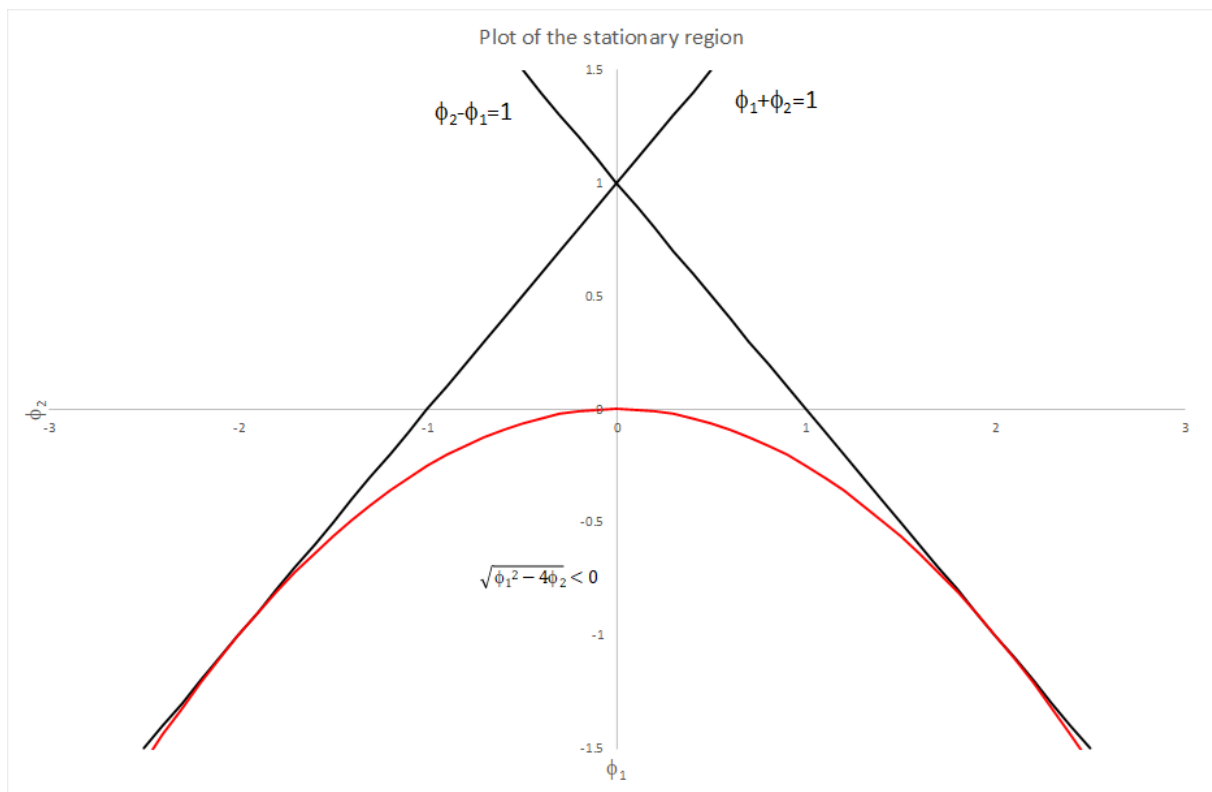
$$w^2 - \phi_1 w - \phi_2 = 0$$

yields 2 solutions

$$w_1, w_2 = \frac{\phi_1 \pm \sqrt{(-\phi_1)^2 + 4\phi_2}}{2}$$

and stationarity requires both roots to be < 1 . You can draw the conditions for the AR(2) process to be stationary or when it has complex roots, see below (Figure 4):

Figure 4



For the AR(p) process to be stationary you require all of the roots to be < 1 (or inside the unit circle)

4 Moving Average (MA) Model

4.1 MA(1)

$$z_t = \theta \varepsilon_{t-1} + \varepsilon_t \quad \varepsilon_t \sim N(0, \sigma^2)$$

In this case the mean can be calculated as:

$$E(z_t) = E(\theta \varepsilon_{t-1} + \varepsilon_t) = E(\theta \varepsilon_{t-1}) + E(\varepsilon_t) = \theta E(\varepsilon_{t-1}) + E(\varepsilon_t) = 0$$

For the variance and the autocovariances we have:

$$V(z_t) = \gamma_0 = V(\theta \varepsilon_{t-1} + \varepsilon_t) = V(\theta \varepsilon_{t-1}) + V(\varepsilon_t) + 2\text{cov}(\theta \varepsilon_{t-1}, \varepsilon_t) = \theta^2 \sigma^2 + \sigma^2 = (1 + \theta^2) \sigma^2$$

$$\text{cov}(z_t, z_{t-1}) = \gamma_1 = E[(\theta \varepsilon_{t-1} + \varepsilon_t)(\theta \varepsilon_{t-2} + \varepsilon_{t-1})] = E[\theta^2 \varepsilon_{t-1} \varepsilon_{t-2} + \theta \varepsilon_{t-1}^2 + \theta \varepsilon_{t-2} \varepsilon_t + \varepsilon_{t-1} \varepsilon_t] = \theta \sigma^2$$

$$\text{cov}(z_t, z_{t-2}) = \gamma_2 = E[(\theta \varepsilon_{t-1} + \varepsilon_t)(\theta \varepsilon_{t-3} + \varepsilon_{t-2})] = E[\theta^2 \varepsilon_{t-1} \varepsilon_{t-3} + \theta \varepsilon_{t-1} \varepsilon_{t-2} + \theta \varepsilon_{t-3} \varepsilon_t + \varepsilon_{t-2} \varepsilon_t] = 0$$

In which case:

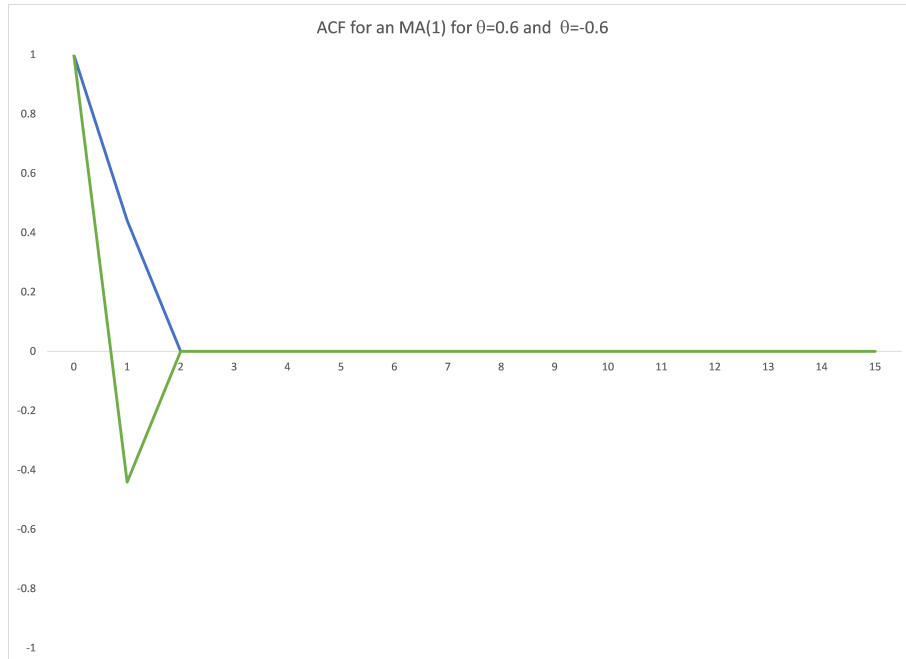
$$\rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{\theta \sigma^2}{(1 + \theta^2) \sigma^2} = \frac{\theta}{(1 + \theta^2)}$$

$$\rho_2 = \frac{\gamma_2}{\gamma_0} = \frac{0}{(1 + \theta^2) \sigma^2} = 0$$

$$\rho_j = 0 \quad j > 1$$

see Figure 5 for the theoretical ACF of an MA(1). (Appendix A Figure A4 provides a sample plot of an MA(1) process and the associated sample ACF and PACF)

Figure 5



Similarly to the fact that we can write an AR model as an infinite MA process, the MA(1) can be written as an infinite AR process:

$$\begin{aligned} z_t &= \theta L \varepsilon_t + \varepsilon_t = (1 + \theta L) \varepsilon_t \\ z_t (1 + \theta L)^{-1} &= \varepsilon_t \\ \text{as } (1 + \theta L)^{-1} &= 1 - \theta L + \theta^2 L^2 - \theta^3 L^3 + \dots \\ z_t &= \theta z_{t-1} - \theta^2 z_{t-2} + \theta^3 z_{t-3} - \theta^4 z_{t-4} + \dots + \varepsilon_t \end{aligned}$$

This ability to write an MA process as an infinite AR process is known as invertibility.

4.2 MA(2)

$$z_t = \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t \quad \varepsilon_t \sim N(0, \sigma^2)$$

In this case the mean can be calculated as:

$$E(z_t) = E(\theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t) = E(\theta_1 \varepsilon_{t-1}) + E(\theta_2 \varepsilon_{t-2}) + E(\varepsilon_t) = \theta_1 E(\varepsilon_{t-1}) + \theta_2 E(\varepsilon_{t-2}) + E(\varepsilon_t) = 0$$

For the variance and the autocovariances we have:

$$V(z_t) = \gamma_0 = V(\theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t) = V(\theta_1 \varepsilon_{t-1}) + V(\theta_2 \varepsilon_{t-2}) + V(\varepsilon_t) = (1 + \theta_1^2 + \theta_2^2) \sigma^2$$

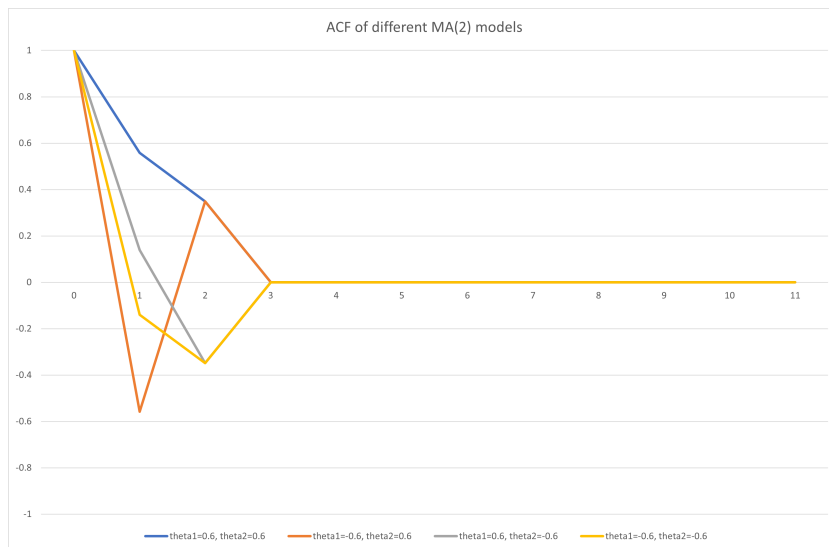
$$\text{cov}(z_t, z_{t-1}) = \gamma_1 = E[(\theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t)(\theta_1 \varepsilon_{t-2} + \theta_2 \varepsilon_{t-3} + \varepsilon_{t-1})] = (\theta_1 + \theta_1 \theta_2) \sigma^2$$

$$\text{cov}(z_t, z_{t-2}) = \gamma_2 = E[(\theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t)(\theta_1 \varepsilon_{t-3} + \theta_2 \varepsilon_{t-4} + \varepsilon_{t-2})] = \theta_2 \sigma^2$$

$$\text{cov}(z_t, z_{t-3}) = \gamma_3 = E[(\theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t)(\theta_1 \varepsilon_{t-4} + \theta_2 \varepsilon_{t-5} + \varepsilon_{t-3})] = 0$$

(see Figure 6 for theoretical examples of the ACFs for an MA(2) process)

Figure 6



4.3 MA(q)

$$z_t = \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q} + \varepsilon_t \quad \varepsilon_t \sim N(0, \sigma^2)$$

In this case the mean can be calculated as:

$$E(z_t) = 0$$

For the variance and the autocovariances we have:

$$\begin{aligned} V(z_t) &= \gamma_0 = (1 + \theta_1^2 + \theta_2^2 + \dots + \theta_q^2) \sigma^2 \\ \text{cov}(z_t, z_{t-1}) &= \gamma_1 = (\theta_1 + \theta_1 \theta_2 + \dots + \theta_{q-1} \theta_q) \sigma^2 \\ \text{cov}(z_t, z_{t-2}) &= \gamma_2 = (\theta_2 + \theta_1 \theta_3 + \dots + \theta_{q-2} \theta_q) \sigma^2 \\ \text{cov}(z_t, z_{t-3}) &= \gamma_3 = (\theta_3 + \theta_1 \theta_4 + \dots + \theta_{q-3} \theta_q) \sigma^2 \\ &\vdots \\ \text{cov}(z_t, z_{t-q}) &= \gamma_q = \theta_q \sigma^2 \\ \text{cov}(z_t, z_{t-j}) &= \gamma_j = 0 \quad j > q \end{aligned}$$

4.4 Calculating Roots

The roots of an MA(q) are obtained by solving the q^{th} order polynomial:

$$w^q + \theta_1 w^{q-1} + \theta_2 w^{q-2} + \dots + \theta_q = 0$$

For invertibility, that is, writing an MA(q) process as an infinite AR process (we require that all roots be < 1). MA processes are by definition stationary (and this hold irrespective of the size of the θ_j . coefficients).

5 ARMA Model

5.1 ARMA(1,1)

$$z_t = \phi z_{t-1} + \theta \varepsilon_{t-1} + \varepsilon_t \quad \varepsilon_t \sim N(0, \sigma^2) \quad (9)$$

We can always write equation (9) using the lag operator (L) as:

$$z_t = \phi L z_t + \theta \varepsilon_{t-1} + \varepsilon_t \Rightarrow z_t(1 - \phi L) = (1 + \theta L)\varepsilon_t \Rightarrow z_t = (1 - \phi L)^{-1}(1 + \theta L)\varepsilon_t$$

Now $(1 - \phi L)^{-1} = 1 + \phi L + \phi^2 L^2 + \phi^3 L^3 + \dots$, in which case:

$$\begin{aligned} z_t &= (1 + \phi L + \phi^2 L^2 + \phi^3 L^3 + \dots)(1 + \theta L)\varepsilon_t \\ &= \varepsilon_t + \phi \varepsilon_{t-1} + \phi^2 \varepsilon_{t-2} + \phi^3 \varepsilon_{t-3} + \dots + \theta \varepsilon_{t-1} + \phi \theta \varepsilon_{t-2} + \phi^2 \theta \varepsilon_{t-3} + \phi^3 \theta \varepsilon_{t-4} + \dots \\ &= \varepsilon_t + (\phi + \theta) \varepsilon_{t-1} + \phi(\phi + \theta) \varepsilon_{t-2} + \phi^2(\phi + \theta) \varepsilon_{t-3} + \phi^3(\phi + \theta) \varepsilon_{t-4} + \dots \end{aligned} \quad (10)$$

Lagging equation (10) by 1:

$$z_{t-1} = \varepsilon_{t-1} + (\phi + \theta) \varepsilon_{t-2} + \phi(\phi + \theta) \varepsilon_{t-3} + \phi^2(\phi + \theta) \varepsilon_{t-4} + \phi^3(\phi + \theta) \varepsilon_{t-5} + \dots \quad (11)$$

This expression enables us to construct the ACF, using equations (9) and (11):

$$\begin{aligned} E[(z_t - \phi z_{t-1})z_t] &= E[(\theta \varepsilon_{t-1} + \varepsilon_t)(\varepsilon_t + (\phi + \theta) \varepsilon_{t-1} + \phi(\phi + \theta) \varepsilon_{t-2} \\ &\quad + \phi^2(\phi + \theta) \varepsilon_{t-3} + \phi^3(\phi + \theta) \varepsilon_{t-4} + \dots)] \\ E[(z_t - \phi z_{t-1})z_t] &= \gamma_0 - \phi \gamma_1 = \sigma^2(1 + \theta(\phi + \theta)) \end{aligned} \quad (12)$$

We can also write using equations (9) and (10) lagged by 1 period

$$\begin{aligned} E[(z_t - \phi z_{t-1})z_{t-1}] &= E[(\theta \varepsilon_{t-1} + \varepsilon_t)(\varepsilon_{t-1} + (\phi + \theta) \varepsilon_{t-2} \\ &\quad + \phi(\phi + \theta) \varepsilon_{t-3} + \phi^2(\phi + \theta) \varepsilon_{t-4} + \phi^3(\phi + \theta) \varepsilon_{t-5} + \dots)] \\ E[(z_t - \phi z_{t-1})z_{t-1}] &= \gamma_1 - \phi \gamma_0 = \sigma^2 \theta \end{aligned} \quad (13)$$

We can also write using equations (9) and (10) lagged by 2 periods

$$\begin{aligned} E[(z_t - \phi z_{t-1})z_{t-2}] &= E[(\theta \varepsilon_{t-1} + \varepsilon_t)(\varepsilon_{t-2} \\ &\quad + (\phi + \theta) \varepsilon_{t-3} + \phi(\phi + \theta) \varepsilon_{t-4} + \phi^2(\phi + \theta) \varepsilon_{t-5} + \phi^3(\phi + \theta) \varepsilon_{t-6} + \dots)] \end{aligned}$$

as none of the terms on the RHS of this equation are the same we have

$$\gamma_2 - \phi \gamma_1 = 0$$

in which case we can say more generally:

$$\gamma_j - \phi\gamma_{j-1} = 0 \quad j \geq 2 \quad (14)$$

Solving (12) and (13) we get:

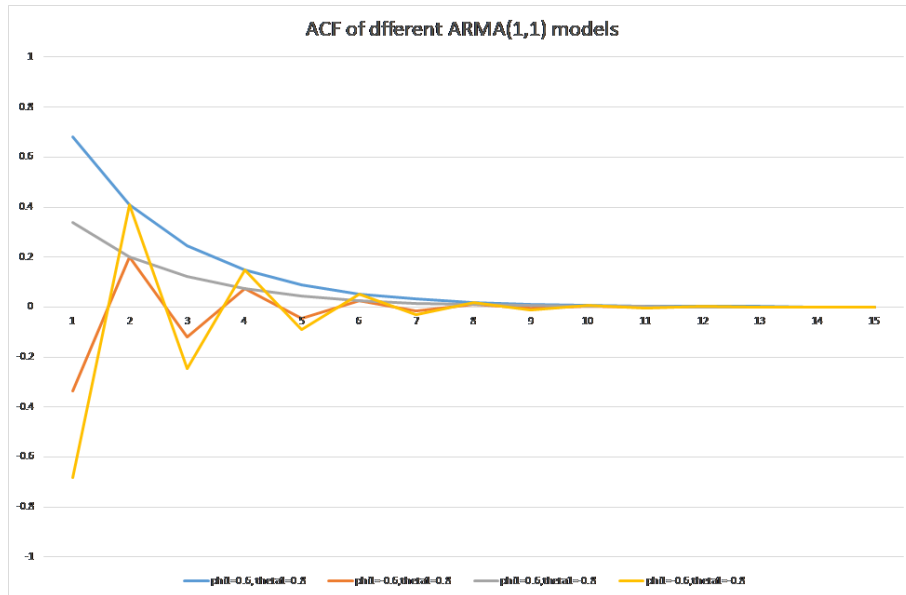
$$\begin{aligned} \gamma_0 - \phi^2\gamma_0 &= \sigma^2(1 + \theta(\phi + \theta)) + \sigma^2\phi\theta \\ \gamma_0 &= \sigma^2(1 + \theta^2 + 2\phi\theta)/(1 - \phi^2) \end{aligned}$$

And hence

$$\gamma_1 = \sigma^2[\theta + \phi(1 + \theta^2 + \phi\theta)]/(1 - \phi^2)$$

All other values can be solved recursively (see Figure 7 for the theoretical ACF of an ARMA(1,1) for different values of ϕ and θ)

Figure 7



6 Partial ACF

Another tool which can be used to distinguish the type of serial correlation present in z_t , is the Partial AutoCorrelation Function (PACF). The PACF is obtained by running a series of regressions:

$$z_t = \rho_{11}z_{t-1} + \eta_t$$

$$z_t = \rho_{21}z_{t-1} + \rho_{22}z_{t-2} + \eta_t$$

$$z_t = \rho_{31}z_{t-1} + \rho_{32}z_{t-2} + \rho_{33}z_{t-3} + \eta_t$$

$$\vdots$$

$$z_t = \rho_{j1}z_{t-1} + \rho_{j2}z_{t-2} + \rho_{j3}z_{t-3} + \dots + \rho_{jj}z_{t-j} + \eta_t$$

And plotting $\rho_{11}, \rho_{22}, \dots, \rho_{jj}$ against j . Each of the parameters is picking up the partial effect of z_{t-j} on z_t , holding constant the effect of $z_{t-1}, z_{t-2}, \dots, z_{t-j+1}$.

Note:

Appendix A contains plots of data and the actual autocorrelation function along with the PACF and the actual series for each of a White Noise process, an AR(1) process (with $\phi = 0.5$ and $\phi = 0.9$) and an MA(1) process. It is possible to calculate confidence intervals around these autocorrelation estimators as:

$$V(\hat{\rho}_1) = 1/n$$

$$V(\hat{\rho}_j) = \frac{1}{n} \left\{ 1 + 2 \sum_{k=1}^{j-1} \hat{\rho}_k^2 \right\}$$

And Stata/R reports the 95% confidence interval around the estimates of the correlation.

The discussion so far has been based on the presumption that we are wanting to look at the ACF of some series, z_t . But we might be interested in looking for the presence of serial correlation in the error term of some regression (like equation (4) in handout 2). As we might be interested in investigating whether it is reasonable to assume: $\text{cov}(\varepsilon_t, \varepsilon_s) = 0$, $t \neq s$, while ε_t is unobserved, we could look at the correlation in the residuals, e_t , as a proxy.

7 Properties of Estimators

In the 2-variable regression model

$$y_t = \beta_0 + \beta_1 x_t + \varepsilon_t$$

where, $E(\varepsilon_t) = 0$, $V(\varepsilon_t) = \sigma^2$, $E(\varepsilon_t \varepsilon_s) = \gamma_{|t-s|} \neq 0$ for $t \neq s$. Following handout 1 (Term 1) it can be shown that

$$b_1 = \beta_1 + \frac{\sum_{t=1} (x_t - \bar{x}) \varepsilon_t}{\sum_{t=1} (x_t - \bar{x})^2} = \beta_1 + \sum_{t=1} \omega_t \varepsilon_t \quad (15)$$

where

$$\omega_t = \frac{(x_t - \bar{x})}{\sum_{t=1} (x_t - \bar{x})^2}$$

7.1 Unbiasedness

Taking expectations of equation (15)

$$E(b_1) = E(\beta_1) + E\left(\sum_{t=1} \omega_t \varepsilon_t\right)$$

as β_1 is a constant, $E(\beta_1) = \beta_1$

$$E(b_1) = \beta_1 + E\left(\sum_{t=1} \omega_t \varepsilon_t\right) = \beta_1 + E(\omega_1 \varepsilon_1 + \omega_2 \varepsilon_2 + \dots + \omega_n \varepsilon_n)$$

Now as ω_t is made up of x_t , and as we are assuming $E(\varepsilon_t | x_1, \dots, x_T) = 0$, then:

$$E(b_1) = \beta_1$$

Therefore the slope coefficient is unbiased estimator.

7.2 Variances

Using equation (15) again, we have that

$$V(b_1) = E[(\hat{\beta}_1 - E(b_1))^2] = E[(\hat{\beta}_1 - \beta_1)^2] = E\left[\left(\sum_{t=1} \omega_t \varepsilon_t\right)^2\right]$$

$$\begin{aligned}
V(b_1) &= E[(\omega_1\varepsilon_1 + \omega_2\varepsilon_2 + \dots + \omega_n\varepsilon_n)^2] = E(\omega_1^2\varepsilon_1^2 + \omega_2^2\varepsilon_2^2 + \dots + \omega_n^2\varepsilon_n^2 \\
&\quad + 2\omega_1\omega_2\varepsilon_1\varepsilon_2 + 2\omega_1\omega_3\varepsilon_1\varepsilon_3 + \dots + 2\omega_1\omega_n\varepsilon_1\varepsilon_n \\
&\quad + 2\omega_2\omega_3\varepsilon_2\varepsilon_3 + \dots + 2\omega_2\omega_n\varepsilon_2\varepsilon_n \\
&\quad + \dots + 2\omega_{n-1}\omega_n\varepsilon_{n-1}\varepsilon_n)
\end{aligned}$$

As the ω_t terms are exogeneous these will be brought outside the expectation operator, that is

$$\begin{aligned}
V(\hat{\beta}_1) &= \omega_1^2 E(\varepsilon_1^2) + \omega_2^2 E(\varepsilon_2^2) + \dots + \omega_n^2 E(\varepsilon_n^2) \\
&\quad + 2\omega_1\omega_2 E(\varepsilon_1\varepsilon_2) + 2\omega_1\omega_3 E(\varepsilon_1\varepsilon_3) + \dots + 2\omega_1\omega_n E(\varepsilon_1\varepsilon_n) \\
&\quad + 2\omega_2\omega_4 E(\varepsilon_2\varepsilon_4) + \dots + 2\omega_2\omega_n E(\varepsilon_2\varepsilon_n) + \dots + 2\omega_{n-1}\omega_n E(\varepsilon_{n-1}\varepsilon_n)
\end{aligned}$$

Now, remember that $V(\varepsilon_t) = E(\varepsilon_t^2) = \sigma^2$ and $cov(\varepsilon_t, \varepsilon_s) = E(\varepsilon_t\varepsilon_s) = \gamma_{|t-s|}$, $t \neq s$. Consequently, the cross-product terms are NOT zero and we have

$$\begin{aligned}
V(b_1) &= \omega_1^2\sigma^2 + \omega_2^2\sigma^2 + \dots + \omega_n^2\sigma^2 + 2\omega_1\omega_2\gamma_1 + 2\omega_1\omega_3\gamma_2 + \dots + 2\omega_1\omega_n\gamma_{n-1} \\
&\quad + 2\omega_2\omega_3\gamma_1 + 2\omega_2\omega_4\gamma_2 + \dots + 2\omega_2\omega_n\gamma_{n-1} + \dots + 2\omega_{n-1}\omega_n\gamma_1 \\
&= \sigma^2 \sum_t \omega_t^2 + 2 \sum_{t=1}^T \sum_{s=t+1}^T \omega_t\omega_s\gamma_{s-t} \\
&= \frac{\sigma^2}{\sum_t (x_t - \bar{x})^2} + 2 \sum_{t=1}^T \sum_{s=t+1}^T \omega_t\omega_s\gamma_{s-t}
\end{aligned}$$

that is, the standard OLS formula

$$\frac{\sigma^2}{\sum_t (x_t - \bar{x})^2}$$

(as used by computer programs, for example Stata and R) gives the incorrect variance estimate, consequently the t-ratios produced by these programmes are wrong!!

More succinctly, we can write

$$V(b_1) = \frac{\sigma_v^2}{\sum_t (x_t - \bar{x})^2} \left[1 + 2 \sum_{j=1}^{n-1} \left(\frac{n-j}{n} \right) \rho_j \right]$$

where $v_t = (x_t - \bar{x})\varepsilon_t$, $\sigma_v^2 = \frac{1}{n} \sum_t (x_t - \bar{x})^2 \varepsilon_t^2$, and $\rho_j = \frac{cov(v_t, v_{t-j})}{var(v_t)}$ and this is the Newey-West's Heteroscedastic Autocorrelation Consistent Standard Errors (HACSE), which is both consistent in the presence of heteroscedasticity and serial correlation.

8 Serial Correlation with y_{t-1}

Consider the simple situation in which we have (although the result holds for more complicated models)

$$y_t = \beta_0 + \beta_1 y_{t-1} + u_t \quad (16)$$

and while u_t satisfies all of our standard classical linear regression model assumptions, we have serial correlation, that is, $cov(u_t, u_{t-1}) \neq 0$ and this might be an AR or MA or ARMA model

Now, OLS estimation of equation (16) yields

$$b_1 = \frac{\sum_{t=1}^n (y_t - \bar{y})(y_{t-1} - \bar{y})}{\sum_{t=1}^n (y_{t-1} - \bar{y})^2}$$

substituting out y_t using equation (16) we have

$$b_1 = \beta_1 + \sum_{t=1}^n \omega_t u_t = \beta_1 + \frac{\sum_{t=1}^n (y_{t-1} - \bar{y}) u_t}{\sum_{t=1}^n (y_{t-1} - \bar{y})^2}$$

$$E(b_1) = \beta_1 + \frac{cov(y_{t-1}, u_t)}{Var(y_{t-1})} \quad (17)$$

Now for unbiasedness we need $cov(y_{t-1}, u_t) = 0$, however, from (16) lagged one period we have

$$y_{t-1} = \beta_0 + \beta_1 y_{t-2} + u_{t-1},$$

that is, $cov(y_{t-1}, u_{t-1}) \neq 0$, but we know that $cov(u_t, u_{t-1}) \neq 0$, therefore $cov(y_{t-1}, u_t) \neq 0$ by definition. Consequently, the second term in equation (17) will never be zero, even as the sample size approaches infinity. Therefore the OLS estimate, when we have a lagged dependent variable and serial correlation, is both BIASED and INCONSISTENT.

Appendix D (E) show the Stata (R) code for a Monte Carlo simulation in which we show that the OLS estimator of the lagged dependent variable in which the errors are serially correlated does not become equal to the true parameter even as the sample size gets very big.

9 Detecting Serial Correlation

9.1 Breusch-Godfrey Test

Consider the general linear regression model

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \dots + \beta_k x_{kt} + u_t \quad (18)$$

where $\text{cov}(u_t, u_s) = E(u_t u_s) \neq 0$, but all of the other CLRM assumptions are satisfied. Now assuming we know the form of the serial correlation is

$$u_t = \delta_0 + \delta_1 u_{t-1} + \dots + \delta_p u_{t-p} + \varepsilon_t$$

where ε_t is some well-behaved error process (satisfying CLRM assumptions). The null hypothesis corresponds to the CLRM assumption of no serial correlation, that is, under $H_0 : \delta_1 = \delta_2 = \dots = \delta_p = 0 \Rightarrow u_t = \delta_0 + \varepsilon_t$, in which case the error term does not depend up on anything and under $H_1 : \text{Any } \delta_j \neq 0$, the error term depends up on previous error terms and is therefore serially correlated.

Unfortunately, u_t is unobserved and we use the residuals (e_t) from equation (18) as a proxy for these, in which case we have

$$e_t = \alpha + \beta_1 x_{1t} + \beta_2 x_{2t} + \dots + \beta_k x_{kt} + \delta_1 e_{t-1} + \dots + \delta_p e_{t-p} + \xi_t \quad (19)$$

although the null and alternative hypotheses (noted above) remain unaltered. The variables $x_{1t}, x_{2t}, \dots, x_{kt}$ are included to correct for degrees of freedom. The null hypothesis is tested via the usual F-test ($= F_{p, T-p-(k+1)}$) or as a Lagrange Multiplier test $LM = TR^2 \sim \chi_p^2$ where T is the number of observations used to estimate equation (19) (this is the Breusch-Godfrey test).

The only issue in the estimation of equation (19) here is the choice of p . One generally uses

Periodicity	p
Annual	1
Quarterly	4
Monthly	12

In all cases you are assuming a form for the serial correlation, which is, in general, unknown, consequently the power of this test will tend to be LOW.

9.2 Durbin Watson Statistic

The Durbin-Watson statistic is an alternative test for detect serial correlation in the error term and is calculated as:

$$DW = \frac{\sum_{t=2}^n (e_t - e_{t-1})^2}{\sum_{t=1}^n e_t^2} = \frac{\sum_t e_t^2 + \sum_t e_{t-1}^2 - 2 \sum_t e_t e_{t-1}}{\sum_{t=1}^n e_t^2}$$

$$DW = 1 + \frac{\sum_t e_{t-1}^2}{\sum_t e_t^2} - 2 \frac{\sum_t e_t e_{t-1}}{\sum_t e_t^2}$$

Now as $\sum_t e_{t-1}^2 \approx \sum_t e_t^2$ we have

$$DW = 2 - 2 \frac{\sum_t e_t e_{t-1}}{\sum_t e_t^2} = 2(1 - \phi)$$

where ϕ is the OLS coefficient from a regression of e_t on e_{t-1} .

ϕ	DW
=0	2
> 0 ($\phi \rightarrow 1$)	$0 < DW < 2$ (DW $\rightarrow 0$)
< 0 ($\phi \rightarrow -1$)	$2 < DW < 4$ (DW $\rightarrow 4$)

The null and alternative hypotheses are written as $H_0 : \phi = 0$ and $H_1 : \phi > (<) 0$

0	d_L	d_U	2	4— d_U	4— d_L	4
Reject H_0 $\rho > 0$	Inconclusive region	Do Not Reject H_0	Do Not Reject H_0	Inconclusive region	Reject H_0 $\rho < 0$	

Including a lagged dependent variable makes the DW test biased towards 2 (i.e. biased towards H_0). If you have a lagged dependent variable a better test than the DW test is Durbin's h test:

$$h = \hat{\phi} \sqrt{\frac{n}{1 - n s_{\hat{\beta}_k}^2}} \sim N(0, 1)$$

where $s_{\hat{\beta}_k}^2$ = the OLS variance estimate on the lagged dependent variable and $\hat{\phi}$ is the estimated 1st autocorrelation coefficient.

BEWARE

Apparent serial correlation could be caused by an omitted relevant variable. Suppose we have a model a true equation given by (T) and an estimated equation given by (F):

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 z_t + \varepsilon_t \quad (\text{T})$$

$$y_t = \delta_0 + \delta_1 x_t + \eta_t \quad (\text{F})$$

It must be the case that $\eta_t = \beta_2 z_t + \varepsilon_t$. In which case we have

$$\begin{aligned} \text{cov}(\eta_t, \eta_{t-1}) &= \text{cov}(\beta_2 z_t + \varepsilon_t, \beta_2 z_{t-1} + \varepsilon_{t-1}) = \beta_2^2 \text{cov}(z_t, z_{t-1}) + \beta_2 \text{cov}(z_t, \varepsilon_{t-1}) \\ &\quad + \beta_2 \text{cov}(\varepsilon_t, z_{t-1}) + \text{cov}(\varepsilon_t, \varepsilon_{t-1}) \end{aligned}$$

Now as ε_t is purely random it must be uncorrelated with everything, in which case

$$\text{cov}(\eta_t, \eta_{t-1}) = \beta_2^2 \text{cov}(z_t, z_{t-1}) \neq 0$$

and we get apparent serial correlation caused by an omitted relevant variable. The most likely omitted relevant variable is either a lagged dependent variable or a lagged explanatory variable. In this case we do NOT actually have serial correlation it is simply a symptom of something else.

Note the misspecification of the dynamics in the equation, would also likely yield apparent serial correlation, although again this is only a symptom of something else. The solution of this is often to include additional lags of the dependent variable or the explanatory variables into your model, where the number of lags to include maybe indicated by the form of the sample ACF.

Appendices

A Plots, ACFs and PACFs from ARMA models

Figure A1: White Noise Process

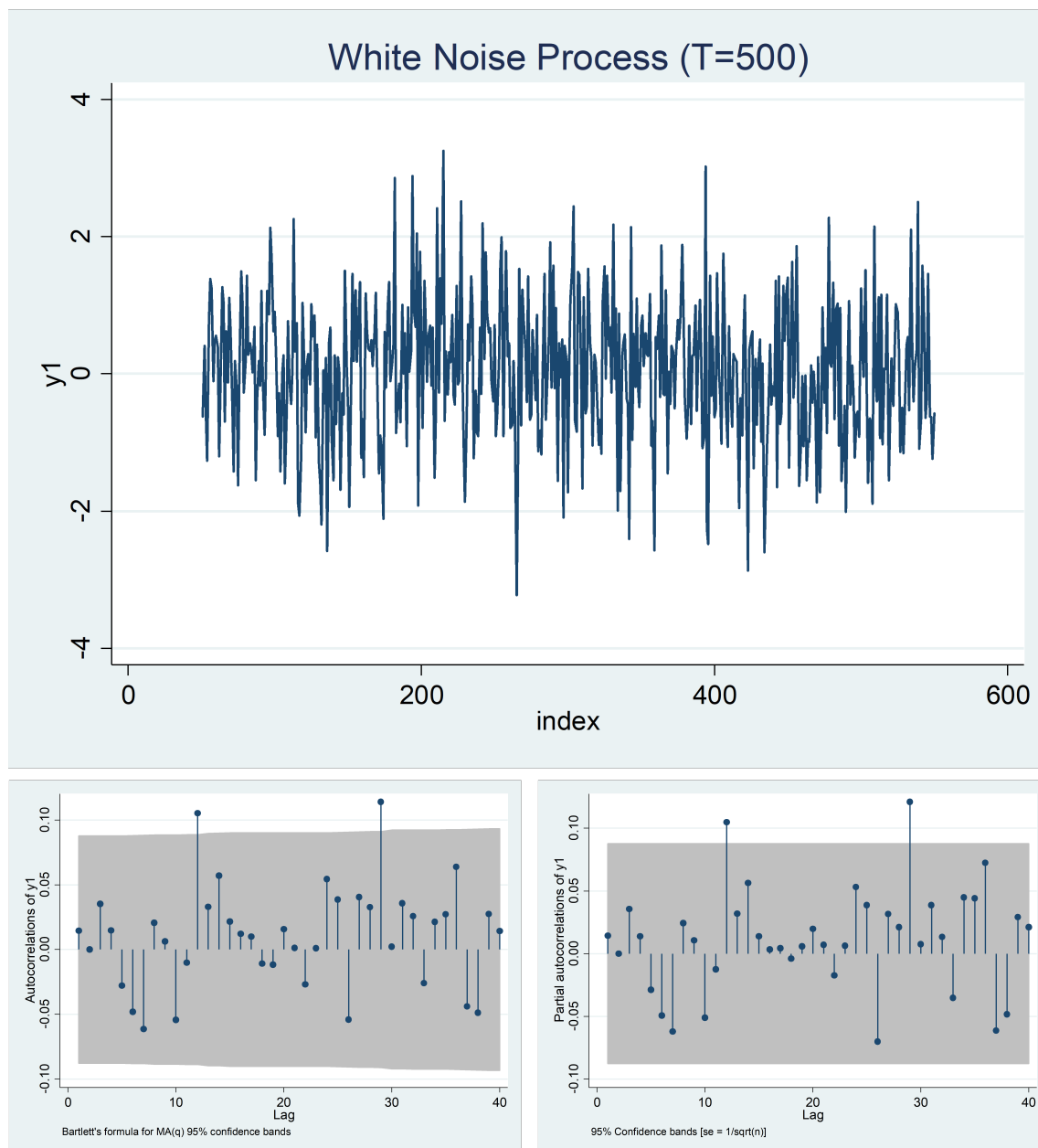


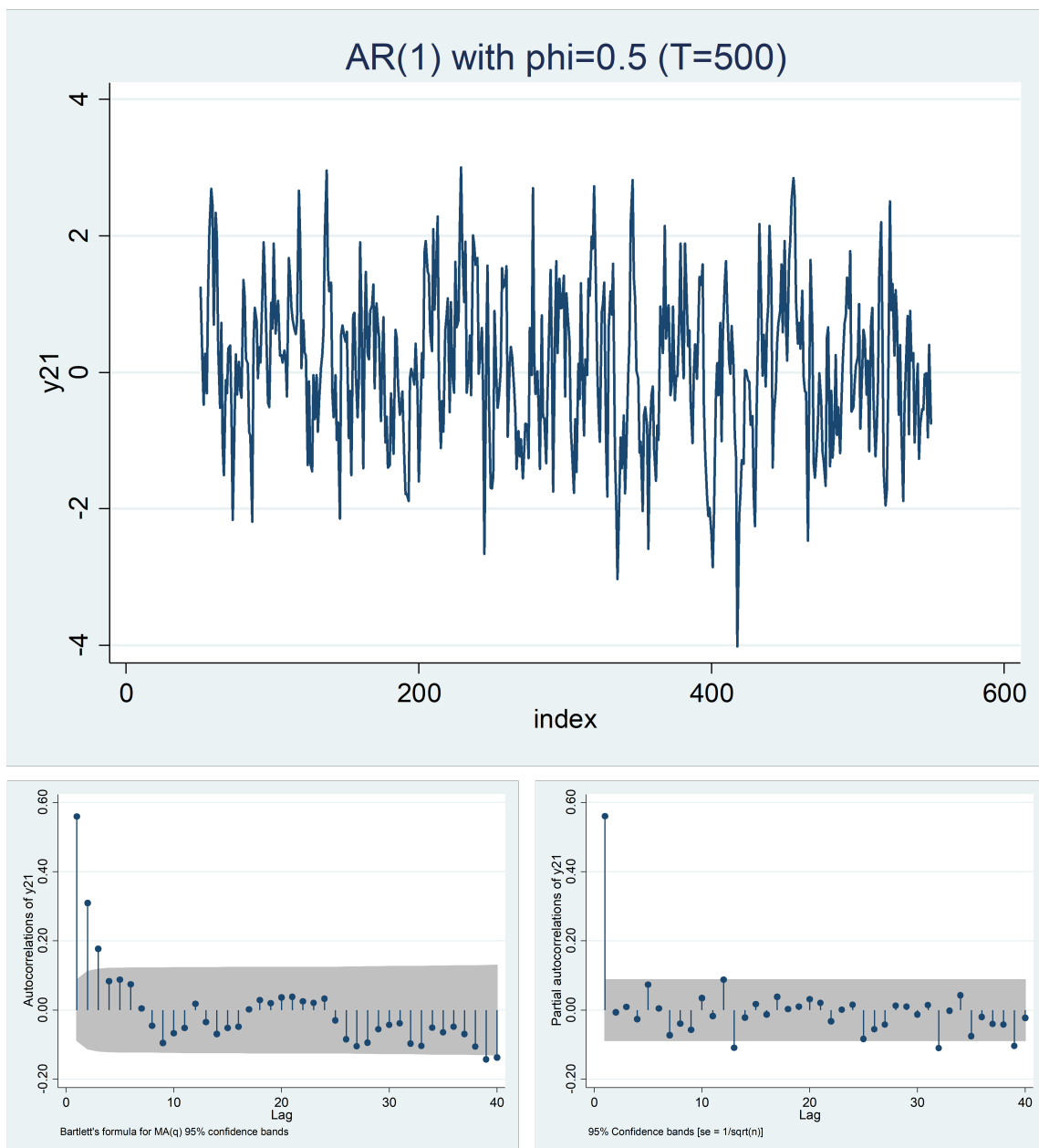
Figure A2: AR(1) $\phi = 0.5$ 

Figure A3: AR(1) $\phi = 0.9$

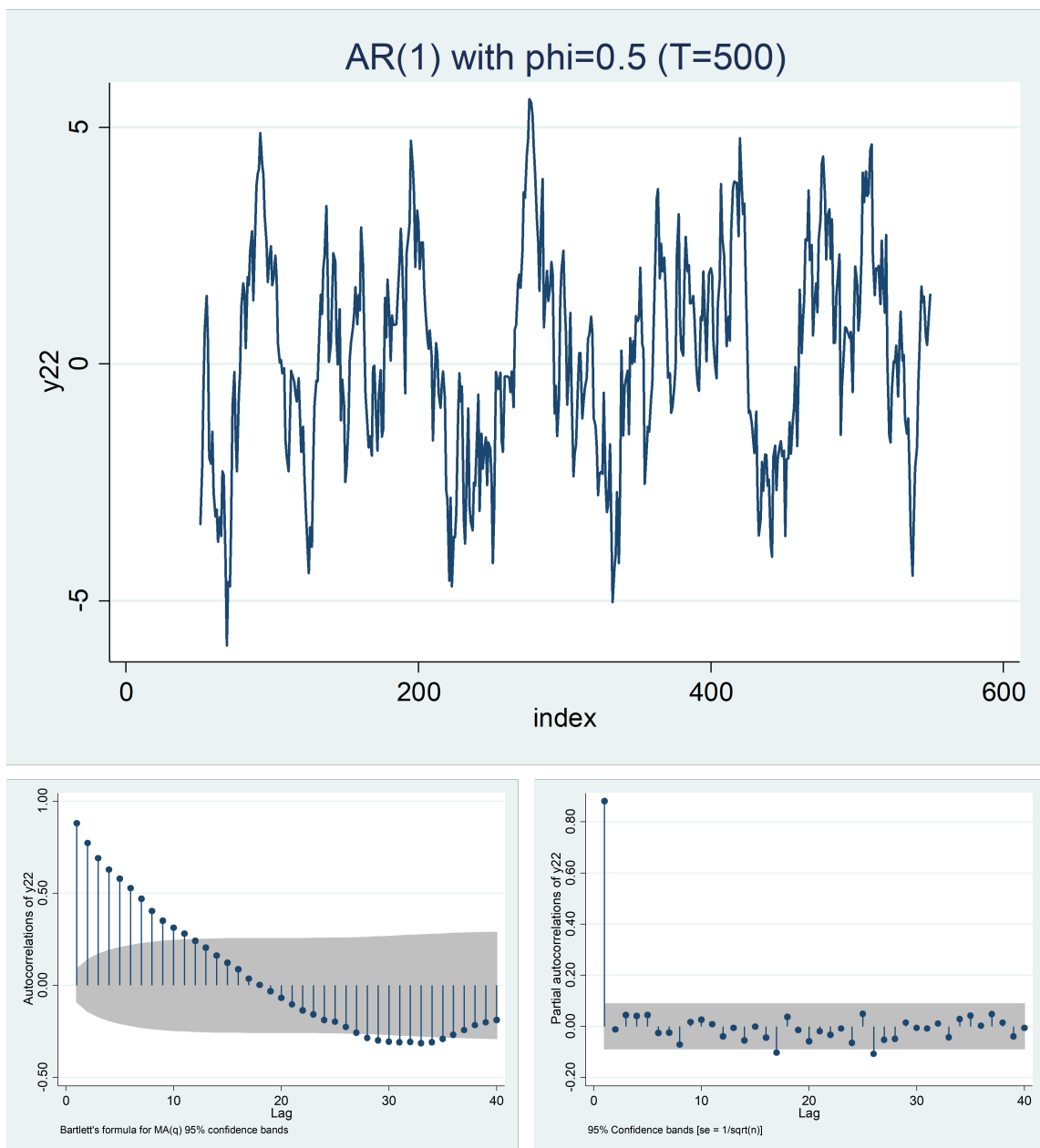
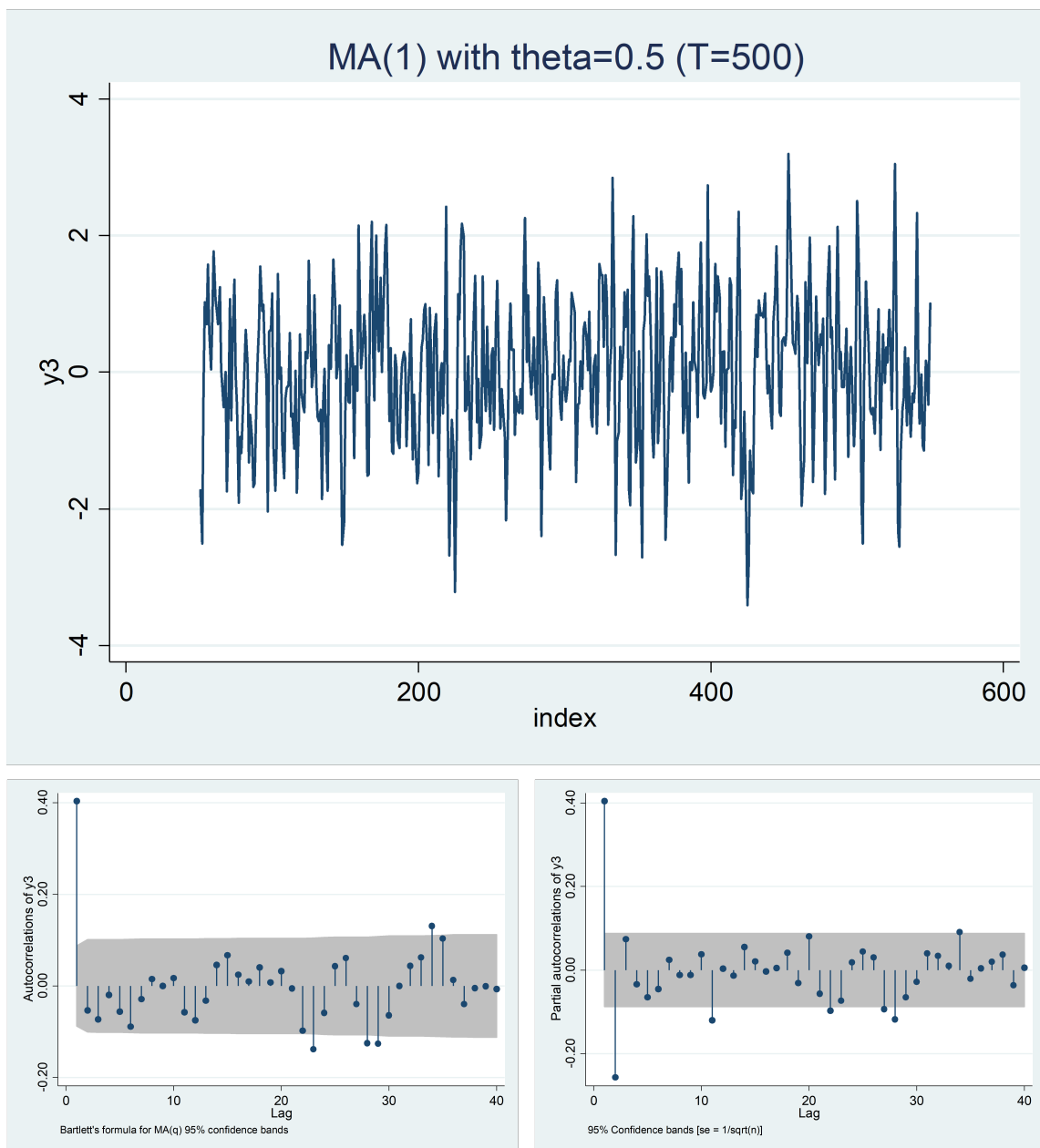


Figure A4: MA(1) $\theta = 0.5$ 

B Stata code to generate ARMA processes

```
clear
/* This sets the start point for random number generator */
set seed 1342
/* The sample size is 500 */
set obs 550
gen index=_n
tsset index

***White Noise process
gen y11=rnormal()
twoway (tsline y11) if(index>50), title(White Noise Process (T=500))
ac y11 if(index>50), lags(10) title(White Noise Process (T=500))
pac y11 if(index>50), lags(10) title(White Noise Process (T=500))

***AR(1) phi=0.6 process
gen y21=0
scalar beta1=0.6
scalar alpha1=0.0
replace y21 = alpha1 + beta1*y21[_n-1] + rnormal() if(index>1)
twoway (tsline y21) if(index>50), title(AR(1) with phi=0.6 (T=500))
ac y21 if(index>50), lags(15) title(AR(1) with phi=0.6 (T=500))
pac y21 if(index>50), lags(15) title(AR(1) with phi=0.6 (T=500))

***AR(1) phi=-0.6 process
gen y22=0
scalar beta2=-0.6
scalar alpha2=0.0
replace y22 = alpha2 + beta2*y22[_n-1] + rnormal() if(index>1)
twoway (tsline y22) if(index>50), title(AR(1) with phi=-0.6 (T=500))
ac y22 if(index>50), lags(15) title(AR(1) with phi=-0.6 (T=500))
pac y22 if(index>50), lags(15) title(AR(1) with phi=-0.6 (T=500))

***AR(2) phi1=0.6, phi2=0.3
gen y31=0
scalar beta1=0.6
scalar beta2=0.3
scalar alpha1=0.0
replace y31 = alpha1 + beta1*y31[_n-1] + beta2*y31[_n-2] + rnormal() if(index>2)
```

```
twoway (tsline y31) if(index>50), title(AR(2) with phi1=0.6 phi2=0.3 (T=500))
ac y31 if(index>50), lags(15) title(AR(2) with phi1=0.6 phi2=0.3 (T=500))
pac y31 if(index>50), lags(15) title(AR(2) with phi1=0.6 phi2=0.3 (T=500))

***AR(2) phi1=1.2, phi2=-0.6
gen y32=0
scalar beta1=1.2
scalar beta2=-0.6
scalar alpha1=0.0
replace y32 = alpha1 + beta1*y32[_n-1] + beta2*y32[_n-2] + rnormal() if(index>2)
twoway (tsline y32) if(index>50), title(AR(2) with phi1=1.2 phi2=-0.6 (T=500))
ac y32 if(index>50), lags(15) title(AR(2) with phi1=1.2 phi2=-0.6 (T=500))
pac y32 if(index>50), lags(15) title(AR(2) with phi1=1.2 phi2=-0.6 (T=500))

***AR(2) phi1=-1.4, phi2=-0.6
gen y33=0
scalar beta1=-1.4
scalar beta2=-0.6
scalar alpha1=0.0
replace y33 = alpha1 + beta1*y33[_n-1] + beta2*y33[_n-2] + rnormal() if(index>2)
twoway (tsline y33) if(index>50), title(AR(2) with phi1=-1.4 phi2=-0.6 (T=500))
ac y33 if(index>50), lags(15) title(AR(2) with phi1=-1.4 phi2=-0.6 (T=500))
pac y33 if(index>50), lags(15) title(AR(2) with phi1=-1.4 phi2=-0.6 (T=500))

***AR(2) phi1=0.7, phi2=-0.8
gen y34=0
scalar beta1=0.7
scalar beta2=-0.8
scalar alpha1=0.0
replace y34 = alpha1 + beta1*y34[_n-1] + beta2*y34[_n-2] + rnormal() if(index>2)
twoway (tsline y34) if(index>50), title(AR(2) with phi1=0.7 phi2=-0.8 (T=500))
ac y34 if(index>50), lags(15) title(AR(2) with phi1=0.7 phi2=-0.8 (T=500))
pac y34 if(index>50), lags(15) title(AR(2) with phi1=0.7 phi2=-0.8 (T=500))

*** MA(1) theta=0.6 process
gen epsilon=rnormal()
gen y41=0
scalar theta=0.6
replace y41= theta*epsilon[_n-1]+epsilon[_n] if(index>1)
```

```
twoway (tsline y41) if(index>50), title(MA(1) with theta=0.6 (T=500))
ac y41 if(index>50), lags(7) title(MA(1) with theta=0.6 (T=500))
pac y41 if(index>50), lags(7) title(MA(1) with theta=0.6 (T=500))
drop epsilon

*** MA(1) theta=-0.6 process
gen epsilon=rnormal()
gen y42=0
scalar theta=-0.6
replace y42= theta*epsilon[_n-1]+epsilon[_n] if(index>1)
twoway (tsline y42) if(index>50), title(MA(1) with theta=-0.6 (T=500))
ac y42 if(index>50), lags(7) title(MA(1) with theta=-0.6 (T=500))
pac y42 if(index>50), lags(7) title(MA(1) with theta=-0.6 (T=500))
drop epsilon

***ARMA(1,1) phi1=0.8, theta1=0.6
gen y51=0
gen epsilon=rnormal()
scalar beta1=0.8
scalar theta=0.6
scalar alpha1=0.0
replace y51 = alpha1 + beta1*y51[_n-1] + theta*epsilon[_n-1]+epsilon[_n] if(index>1)
twoway (tsline y51) if(index>50), title(ARMA(1,1) with phi1=0.8 theta=0.6 (T=500))
ac y51 if(index>50), lags(15) title(ARMA(1,1) with phi1=0.8 theta=0.6 (T=500))
pac y51 if(index>50), lags(15) title(ARMA(1,1) with phi1=0.8 theta=0.6 (T=500))
drop epsilon

***ARMA(1,1) phi1=0.8, theta1=-0.6
gen y52=0
gen epsilon=rnormal()
scalar beta1=0.8
scalar theta=-0.6
scalar alpha1=0.0
replace y52 = alpha1 + beta1*y52[_n-1] + theta*epsilon[_n-1]+epsilon[_n] if(index>1)
twoway (tsline y52) if(index>50), title(ARMA(1,1) with phi1=0.8 theta=-0.6 (T=500))
ac y52 if(index>50), lags(15) title(ARMA(1,1) with phi1=0.8 theta=-0.6 (T=500))
pac y52 if(index>50), lags(15) title(ARMA(1,1) with phi1=0.8 theta=-0.6 (T=500))
drop epsilon
```

```
***ARMA(1,1) phi1=-0.8, theta1=0.6
gen y53=0
gen epsilon=rnormal()
scalar beta1=-0.8
scalar theta=0.6
scalar alpha1=0.0
replace y53 = alpha1 + beta1*y53[_n-1] + theta*epsilon[_n-1]+epsilon[_n] if(index>1)
twoway (tsline y53) if(index>50), title(ARMA(1,1) with phi1=-0.8 theta=0.6 (T=500))
ac y53 if(index>50), lags(15) title(ARMA(1,1) with phi1=-0.8 theta=0.6 (T=500))
pac y53 if(index>50), lags(15) title(ARMA(1,1) with phi1=-0.8 theta=0.6 (T=500))
drop epsilon

***ARMA(1,1) phi1=0.8, theta1=0.6
gen y54=0
gen epsilon=rnormal()
scalar beta1=-0.6
scalar theta=-0.6
scalar alpha1=0.0
replace y54 = alpha1 + beta1*y54[_n-1] + theta*epsilon[_n-1]+epsilon[_n] if(index>1)
twoway (tsline y54) if(index>50), title(ARMA(1,1) with phi1=-0.6 theta=-0.6 (T=500))
ac y54 if(index>50), lags(15) title(ARMA(1,1) with phi1=-0.6 theta=-0.6 (T=500))
pac y54 if(index>50), lags(15) title(ARMA(1,1) with phi1=-0.6 theta=-0.6 (T=500))
drop epsilon
```

C R code to generate ARMA processes

```
library(forecast)
library(dplyr)
library(tidyr)
library(urca)

# This sets the start point for random number generator */
set.seed(seed=1342)
#The sample size is 500 */
n <- 550
t <- seq(1, n)

#White Noise process
y11 <- arima.sim(model = list(order = c(0, 0, 0)), n = n, mean = 0)
plot(y11[t>50], type="l")
title("White Noise Process (T=500)")
acf_y11 <- acf(y11[t>50], plot = F)
plot (acf_y11, main = "ACF White Noise Process (T=500)")
pacf_y11 <- pacf(y11[t>50], plot = F)
plot (pacf_y11, main = "PACF White Noise Process (T=500)")

#AR(1) phi=0.6 process
y21 <- arima.sim(model = list(order = c(1, 0, 0), ar=0.6), n = n, mean = 0)
plot(y21[t>50], type="l",
      main = expression(paste("AR(1), ", phi, "=0.6 (T=500)")))
acf_y21 <- acf(y21[t>50], plot = F)
plot (acf_y21,
      main = expression(paste("ACF for AR(1), ", phi, "=0.6 (T=500)")))
pacf_y21 <- pacf(y21[t>50], plot = F)
plot (pacf_y21,
      main = expression(paste("PACF for AR(1), ", phi, "=0.6 (T=500)")))

#AR(1) phi=-0.6 process
y22 <- arima.sim(model = list(order = c(1, 0, 0), ar=-0.6), n = n, mean = 0)
plot(y22[t>50], type="l",
      main = expression(paste("AR(1), ", phi, "=-0.6 (T=500)")))
acf_y22 <- acf(y22[t>50], plot = F)
plot (acf_y22,
      main = expression(paste("ACF for AR(1), ", phi, "=-0.6 (T=500)")))
```

```
pacf_y22 <- pacf(y22[t>50], plot = F)
plot (pacf_y22,
      main = expression(paste("PACF for AR(1), ", phi, "=-0.6 (T=500)")))

#AR(2) phi1=0.6, phi2=0.3
y31 <- arima.sim(model = list(order = c(2, 0, 0), ar=c(0.6, 0.3)), n = n, mean = 0)
plot (y31[t>50], type="l",
      main = expression(paste("AR(2), ", phi[1], "=0.6, ", phi[2], "=0.3 (T=500)")))
acf_y31 <- acf(y31[t>50], plot = F)
plot (acf_y31,
      main = expression(paste("ACF for AR(2), ", phi[1], "=0.6, ", phi[2], "=0.3 (T=500)")))
pacf_y31 <- pacf(y31[t>50], plot = F)
plot (pacf_y31,
      main = expression(paste("PACF for AR(2), ", phi[1], "=0.6, ", phi[2], "=0.3 (T=500)")))

#AR(2) phi1=1.2, phi2=-0.6
y32 <- arima.sim(model = list(order = c(2, 0, 0), ar=c(1.2, -0.6)), n = n, mean = 0)
plot (y32[t>50], type="l",
      main = expression(paste("AR(2), ", phi[1], "=1.2, ", phi[2], "=-0.6 (T=500)")))
acf_y32 <- acf(y32[t>50], plot = F)
plot (acf_y32,
      main = expression(paste("ACF for AR(2), ", phi[1], "=1.2, ", phi[2], "=-0.6 (T=500)")))
pacf_y32 <- pacf(y32[t>50], plot = F)
plot (pacf_y32,
      main = expression(paste("PACF for AR(2), ", phi[1], "=1.2, ", phi[2], "=-0.6 (T=500)")))

#AR(2) phi1=-1.4, phi2=-0.6
y33 <- arima.sim(model = list(order = c(2, 0, 0), ar=c(-1.4, -0.6)), n = n, mean = 0)
plot (y33[t>50], type="l",
      main = expression(paste("AR(2), ", phi[1], "=-1.4, ", phi[2], "=-0.6 (T=500)")))
acf_y33 <- acf(y33[t>50], plot = F)
plot (acf_y33,
      main = expression(paste("ACF for AR(2), ", phi[1], "=-1.4, ", phi[2], "=-0.6 (T=500)")))
pacf_y33 <- pacf(y33[t>50], plot = F)
plot (pacf_y33,
      main = expression(paste("PACF for AR(2), ", phi[1], "=-1.4, ", phi[2], "=-0.6 (T=500)")))

#AR(2) phi1=0.7, phi2=-0.8
y34 <- arima.sim(model = list(order = c(2, 0, 0), ar=c(0.7, -0.8)), n = n, mean = 0)
```



```
plot (y34[t>50], type="l",
      main = expression(paste("AR(2), ", phi[1], "=0.7, ", phi[2], "=-0.8 (T=500)")))
acf_y34 <- acf(y34[t>50], plot = F)
plot (acf_y34,
      main = expression(paste("ACF for AR(2), ", phi[1], "=0.7, ", phi[2], "=-0.8 (T=500)")))
pacf_y34 <- pacf(y34[t>50], plot = F)
plot (pacf_y34,
      main = expression(paste("PACF for AR(2), ", phi[1], "=0.7, ", phi[2], "=-0.8 (T=500)")))

#MA(1) theta=0.6 process
y41 <- arima.sim(model = list(order = c(0, 0, 1), ma=c(0.6)), n = n, mean = 0)
plot (y41[t>50], type="l",
      main = expression(paste("MA(1), ", theta, "=0.6 (T=500)")))
acf_y41 <- acf(y41[t>50], plot = F)
plot (acf_y41,
      main = expression(paste("ACF for MA(1), ", theta, "=0.6 (T=500)")))
pacf_y41 <- pacf(y41[t>50], plot = F)
plot (pacf_y41,
      main = expression(paste("PACF for MA(1), ", theta, "=0.6 (T=500)")))

#MA(1) theta=-0.6 process
y42 <- arima.sim(model = list(order = c(0, 0, 1), ma=c(-0.6)), n = n, mean = 0)
plot (y42[t>50], type="l",
      main = expression(paste("MA(1), ", theta, "=-0.6 (T=500)")))
acf_y42 <- acf(y42[t>50], plot = F)
plot (acf_y42,
      main = expression(paste("ACF for MA(1), ", theta, "=-0.6 (T=500)")))
pacf_y42 <- pacf(y42[t>50], plot = F)
plot (pacf_y42,
      main = expression(paste("PACF for MA(1), ", theta, "=-0.6 (T=500)")))

#ARMA(1,1) phi1=0.8, theta1=0.6
y51 <- arima.sim(model = list(order = c(1, 0, 1), ar=0.8, ma=0.6), n = n, mean = 0)
plot (y51[t>50], type="l",
      main = expression(paste("ARMA(1, 1), ", phi, "=0.8, ", theta, "=0.6 (T=500)")))
acf_y51 <- acf(y51[t>50], plot = F)
plot (acf_y51,
      main = expression(paste("ACF for ARMA(1, 1), ", phi, "=0.8, ", theta, "=0.6 (T=500)")))
pacf_y51 <- pacf(y51[t>50], plot = F)
```

```

plot (pacf_y51,
      main = expression(paste("PACF for ARMA(1, 1), ", phi, "=0.8, ", theta, "=0.6 (T=500)"))

#ARMA(1,1) phi1=0.8, theta1=-0.6
y52 <- arima.sim(model = list(order = c(1, 0, 1), ar=0.8, ma=-0.6), n = n, mean = 0)
plot (y52[t>50], type="l",
      main = expression(paste("ARMA(1, 1), ", phi, "=0.8, ", theta, "=-0.6 (T=500)")))
acf_y52 <- acf(y52[t>50], plot = F)
plot (acf_y52,
      main = expression(paste("ACF for ARMA(1, 1), ", phi, "=0.8, ", theta, "=-0.6 (T=500)"))
pacf_y52 <- pacf(y52[t>50], plot = F)
plot (pacf_y52,
      main = expression(paste("PACF for ARMA(1, 1), ", phi, "=0.8, ", theta, "=-0.6 (T=500)"))

#ARMA(1,1) phi1=-0.8, theta1=0.6
y53 <- arima.sim(model = list(order = c(1, 0, 1), ar=-0.8, ma=0.6), n = n, mean = 0)
plot (y53[t>50], type="l",
      main = expression(paste("ARMA(1, 1), ", phi, "=-0.8, ", theta, "=0.6 (T=500)")))
acf_y53 <- acf(y53[t>50], plot = F)
plot (acf_y53,
      main = expression(paste("ACF for ARMA(1, 1), ", phi, "=-0.8, ", theta, "=0.6 (T=500)"))
pacf_y53 <- pacf(y53[t>50], plot = F)
plot (pacf_y53,
      main = expression(paste("PACF for ARMA(1, 1), ", phi, "=-0.8, ", theta, "=0.6 (T=500)"))

#ARMA(1,1) phi1=-0.8, theta1=-0.6
y54 <- arima.sim(model = list(order = c(1, 0, 1), ar=-0.8, ma=-0.6), n = n, mean = 0)
plot (y54[t>50], type="l",
      main = expression(paste("ARMA(1, 1), ", phi, "=-0.8, ", theta, "=-0.6 (T=500)")))
acf_y54 <- acf(y54[t>50], plot = F)
plot (acf_y54,
      main = expression(paste("ACF for ARMA(1, 1), ", phi, "=-0.8, ", theta, "=-0.6 (T=500)"))
pacf_y54 <- pacf(y54[t>50], plot = F)
plot (pacf_y54,
      main = expression(paste("PACF for ARMA(1, 1), ", phi, "=-0.8, ", theta, "=-0.6 (T=500)"))

```

D Monte Carlo: $y_{t-1} + AR$ errors (Stata)

```
clear all
global nobs = 200
global nmc = 1000
set seed 10101
set obs $nobs
gen time = _n
tsset time

scalar sigma = 5 /* sd of errors */
scalar delta = .5 /* coeff for lagged y */
scalar rho = .8 /* autocorrelation in errors */

gen u = 0
gen y = 0

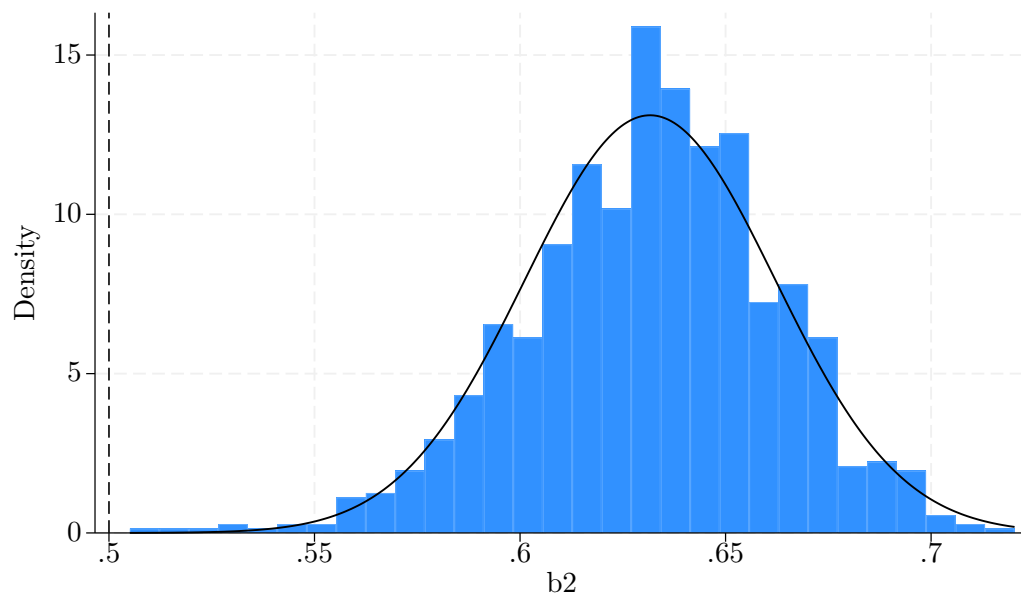
program regLDV, rclass
tempname sim
postfile 'sim' b1 se1 using results, replace
quietly {

forvalues i = 1/$nmc {
* generate errors and samples of y
replace u = rho * L.u + rnormal(0,sigma) in 2/$nobs
replace y = delta * L.y + u in 2/$nobs
reg L(0/1).y
scalar b1 = _b[L.y]
scalar se1 = _se[L.y]
post 'sim' (b1) (se1)
}
}
postclose 'sim'
end

regLDV
use results, clear
su b1 se1
histogram b1, bin(30) normal xline(0.5, lwidth(medthin)) xlabel(0.5(0.05)0.7)
```

A plot of the distribution of the coefficient on the lagged dependent variable can be seen in Figure D1

Figure D1 - Bias in the coefficient on lagged dependent variable with serial correlation



E Monte Carlo: $y_{t-1} + AR$ errors (R)

```
library(ggplot2)
library(tidy)
library(dplyr)

# This sets the start point for random number generator */
set.seed(seed = 1342)
# The sample size is 30
n <- 550
rho <- 0.2
sigma <- 5
delta <- .5
nrep <- 1000
b_1 <- numeric(nrep)

for (j in 1:nrep) {
  y <- 0
  e <- 0
  u2 <- rnorm(n, mean=0, sd=sigma)
  for(i in 2:length(e)){
    e[i] <- rho * e[i-1] + u2[i]
    y[i] <- delta * y[i-1] + e[i]
  }
  ly=lag(y)
  df_x <- data.frame(y, ly)
  ols_fit <- lm(y ~ x + ly, data = df_x)
  b_1[j] <- as.vector(ols_fit$coef[2])
  df_coef <- data.frame(slope= b_1)
}

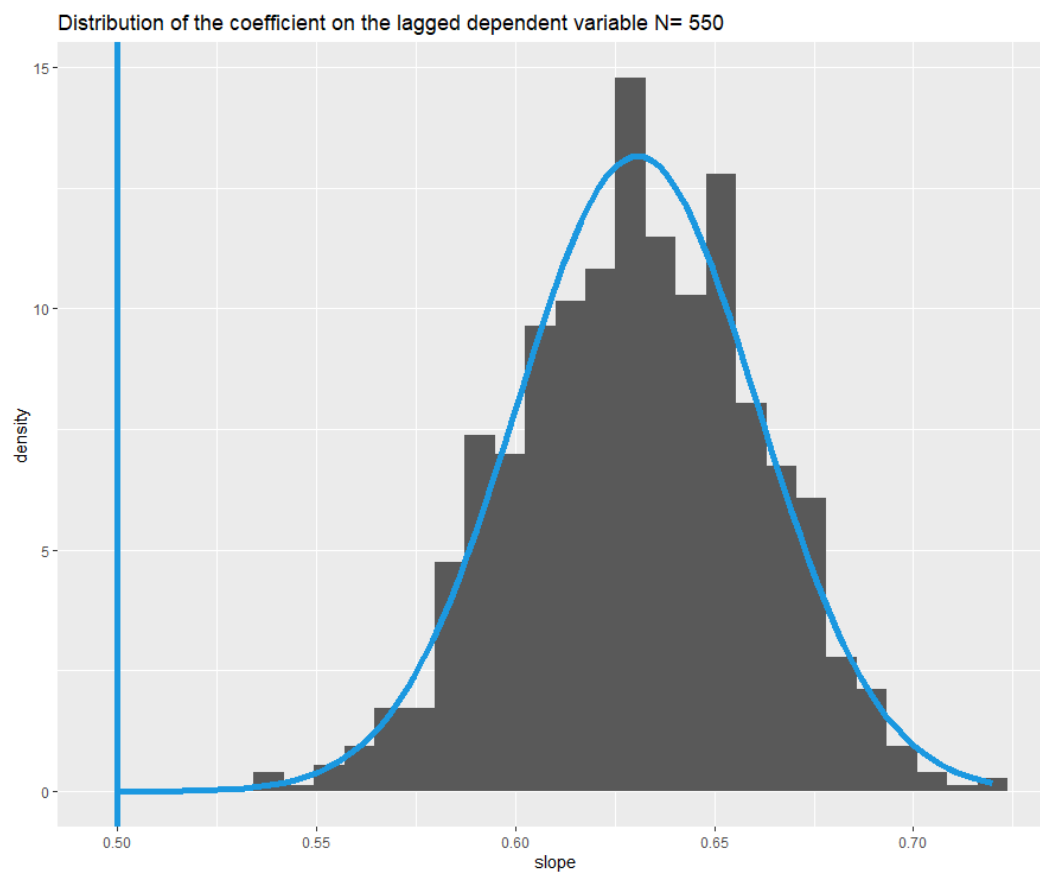
df_coef %>%
  summarise(m_slope = mean(slope),
            sd_slope = sd(slope))

ggplot(df_coef, aes(slope)) +
  geom_histogram(aes(y = ..density..)) +
  stat_function(fun = dnorm,
               args = list(mean = mean(df_coef$slope),
                           sd = sd(df_coef$slope)),
```

```
col = "#1b98e0",  
size = 2) +  
geom_vline(xintercept = delta, color="#1b98e0", lwd =2) +  
ggtitle(paste("Distribution of the coefficient on the lagged dependent variable N=", n))
```

A plot of the distribution of the coefficient on the lagged dependent variable can be seen in Figure E1

Figure E1 - Bias in the coefficient on lagged dependent variable with serial correlation



F Stata Commands for time series models

For quarterly data of the form:

	date	alc	inc
1.	1985Q1	15610	224011
2.	1985Q2	16557	228726
3.	1985Q3	16899	228942
4.	1985Q4	17218	229491
5.	1986Q1	16114	232164
6.	1986Q2	17149	233823
7.	1986Q3	17556	234862
...	...	...	...

```
\begin{verbatim}
/* To declare the series as quarterly data we type:
generate time = quarterly(date, "yq")
format time %tq
tsset time
/* Plot the series of alc from 1990Q1 and name the graph, tsalc */
tsline alc if time>=tq(1990Q1), name(tsalc, replace)
/* Plot the ACF with 20 lags of alc from 1990Q1 and name the graph acalc */
ac alc if time>=tq(1990Q1), lags(20) name(acalc, replace)
/* Plot the PACF with 20 lags of alc from 1990Q1 and name the graph, pacalc */
pac alc if time>=tq(1990Q1), lags(20) name(pacalc, replace)
/* Combine the plots into one graph */
graph combine tsalc acalc pacalc, rows(2) cols(2)
/* Estimating an equation with 1 lag of the dependent variable and 2 of the
explanatory variable over the period from 1990Q1 to 2015Q4 and then test for
serial correlation in the error using the BG test (along with the DW test and
Durbin h-test) and then to plot the ACF and PACF of the residuals */
reg l(0/1).alc l(0/2).gdp if(time>=tq(1990Q1) & time<=tq(2015q4))
predict res, resid
estat bgodfrey, lags(4)
estat dwatson
estat durbinalt
ac res if(time>=tq(1990Q1) \& time<=tq(2015q4)), lags(20)
pac res if(time>=tq(1990Q1) \& time<=tq(2015q4)), lags(20)
/* If you want to estimate the above equation using West's HACSEs with up to 6 lags */
newey l(0/1).alc l(0/2).gdp if(time>=tq(1990Q1) \& time<=tq(2015q4)), lag(6)
```

G R Commands for time series models

#In R you must install the appropriate package. Having installed the package
#you must load the library associated with that package each time you enter R.

#I used the following libraries over Term 2:

```
library(tseries)
library(urca)
library(forecast)
library(ggplot2)
library(dplyr)
library(zoo)
library(xts)
library(ggfortify)
library(dyn)
```

For quarterly data of the form:

	date	alc	inc
1.	1985Q1	15610	224011
2.	1985Q2	16557	228726
3.	1985Q3	16899	228942
4.	1985Q4	17218	229491
5.	1986Q1	16114	232164
6.	1986Q2	17149	233823
7.	1986Q3	17556	234862
...	...	...	...

#In R you must install the appropriate package. Having installed the package
#you must load the library associated with that package each time you enter R.

#I used the following libraries over Term 2:

```
library(tseries)
library(urca)
library(forecast)
library(ggplot2)
library(dplyr)
library(zoo)
library(xts)
library(ggfortify)
library(dyn)
library(lmtest)
library(car)
```

```

# Read in the series as quarterly data:
uk_alc_q <- read_excel("C:/Downloads/uk_alc_q.xlsx")
uk_alc_q <- uk_alc_q %>%
  select(-c(date))
date2 <- seq(as.Date("1985-1-1"), as.Date("2015-4-1"), by = "quarters")
uk_alc_tsq <- xts(uk_alc_q, date2)
# Plot the series of alc from 1990Q1
autoplot(subset(uk_alc_tsa$alc, date2>=as.Date('1990-01-01')), geom = "line") +
  ggtitle("Plot of UK Alcohol consumption") +
  ylab("GDP") + xlab("Year")
# Plot the ACF of alc from 1990Q1
acf_alc <- acf(subset(uk_alc_tsq$alc, date2>=as.Date('1990-1-1')))
# Plot the ACF of alc from 1990Q1
pacf_alc <- pacf(subset(uk_alc_tsq$alc, date2>=as.Date('1990-1-1')))
# Combine the plots into one graph
par(mfrow = c(1, 2))
acf(subset(uk_alc_tsq$alc, date2>=as.Date('1990-1-1')))
pacf(subset(uk_alc_tsq$alc, date2>=as.Date('1990-1-1')))
# Estimating an equation with 1 lag of the dependent variable and 2 of the
# explanatory variable over the period from 1990Q1 to 2015Q4 and then test for
# serial correlation in the error using the BG test (along with the DW test and
# Durbin h-test) and then to plot the ACF and PACF of the residuals */
f2 <- dyn$lm(alc ~ gdp + lag(gdp, 1) + lag(gdp, 2) + lag(alc, 1),
  data=subset(uk_alc_tsq, date2>=as.Date('1989-10-01')))
# This saves the residuals
resids<-f2$resid
resids <- data.frame(resids)
# This does a BG test
bgtest(f2, order=1, data=danish_tsq)
# This does a DW test
durbinWatsonTest(f3)
# Plotting the ACF and PACF of the residuals must create a new date series
# and a new dataframe and from this declare as xts
date3 <- seq(as.Date("1990-01-01"), as.Date("2015-04-01"), by = "quarters")
uk_alc_tsq2 <- xts(resids, order.by=date4)
acf(subset(uk_alc_tsq2$resids, date3>=as.Date('1990Q1')))
pacf(subset(uk_alc_tsq2$resids, date3>=as.Date('1990Q1')))
# If you want to estimate the above equation using West's HACSEs witha up to 6 lags */

```

```
f3 <- dyn$lm(alc ~ lag(alc, 1) + gdp + lag(gdp, 1) + lag(gdp, 2),  
  data=subset(uk_alc_tsq, date2>=as.Date('1989-10-01')))  
coeftest(f3, vcov.=NeweyWest(f3, lag=6, prewhite=FALSE, adjust=TRUE, verbose=TRUE))
```

References

Dougherty, C. (2016). *Introduction to Econometrics*. OUP Catalogue. Oxford University Press.

Stock, J. and Watson, M. W. (2003). *Introduction to Econometrics*. Prentice Hall, New York.

Wooldridge, J. M. (2013). *Introduction to Econometrics*. Cengage.